C13 Skeletal Animation (optional)

CRE132 worksheet

What you'll make

This sample is **optional**. It compares two ways of animating a character. The pixel-art hero from C12 walks **frame by frame**: six separate drawings. The knight walks with **skeletal animation**: one set of body parts moved by **bones**, like a puppet.

You'll look at how the knight's parts are prepared, then rig an arm of your own: give the parts bones with **Sprite Skin**, join the bones into a chain, animate them, and pose the arm with **IK** (inverse kinematics), where you drag the hand and the elbow bends by itself. At the end you'll compare it with two whole knights built the same way. No code.

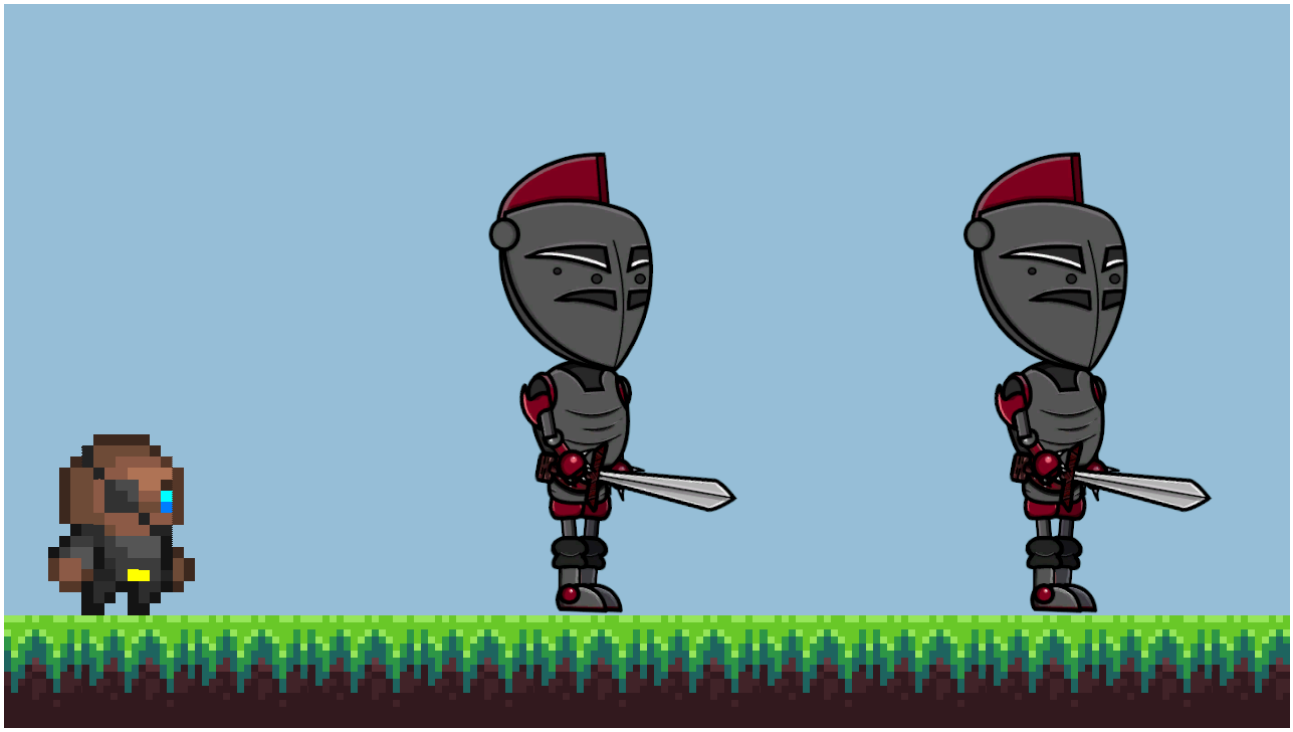


Figure 1: The finished scene: the pixel-art hero on the left and two large armoured knights built from parts.

Before you start

1. Unzip `CRE132_C13_SkeletalAnimation_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_C13_SkeletalAnimation_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Core/C13_SkeletalAnimation`.
3. Press **Play**. **Hero (Frame by Frame)**, on the left, plays C12's walk: a new drawing for every pose. The rest of the grass is empty: that's where your arm goes. Press Play again to stop.

The knight's parts are already in the project, in `Assets/_game/Art/Sprites/Knight/KnightParts.png`, sliced and with their bones. You'll put three of them together into an arm.

1. Frame by frame or skeletal?

- **Frame by frame** needs a new drawing for every pose. It suits small pixel-art characters, which usually look better that way.
- **Skeletal** reuses one set of parts and moves them. New animations cost no new art, and they're smooth at any frame rate. It suits big, smooth, hand-drawn characters like the knight you'll see at the end.

2. The parts and their pivots

Click `Assets/_game/Art/Sprites/Knight/KnightParts.png`. It isn't pixel art, so its settings differ from C01: **Pixels Per Unit 100** and **Filter Mode Bilinear**. Open the **Sprite Editor**. The sheet is sliced into nine parts: `knight_torso`, `knight_head`, `knight_upperarm`, `knight_forearm`, `knight_hand`, `knight_thigh`, `knight_shin`, `knight_foot` and `knight_sword`.

Click `knight_upperarm`: its **Pivot** is **Custom**, placed on the shoulder. Every part has its pivot on its joint (shoulder, elbow, hip, knee…), because that's the point it turns around.

3. The Skinning Editor

Still in the Sprite Editor, open the drop-down at the top left (it says **Sprite Editor**) and choose **Skinning Editor**. Click the upper arm:

- It has one **bone**, drawn from the shoulder to the elbow. Bones are made with **Create Bone**: click at the joint, then at the far end.
- **Auto Geometry** made a **mesh** for the sprite, and **Auto Weights** tied the mesh to the bone, so the picture follows when the bone moves.

All nine parts are already done like this. Each part is rigid, so one bone per part is enough. Close the Sprite Editor without applying any changes (if Unity asks, choose **Revert**).

4. Sprite Skin: giving a part its bone

You'll rig one arm, in the empty space to the right of the hero.

1. Choose **GameObject > Create Empty** from the menu bar. Name it **My Arm** and set its **Position** to 0, 3, 0, near the middle of the camera view.
2. In the Project window, click the small arrow on `KnightParts` to see its parts. Drag `knight_upperarm`, `knight_forearm` and `knight_hand` into the Scene view, then drag each one onto **My Arm** in the Hierarchy so they're its children.
3. Set the **Position** of each of the three parts to 0, 0, 0. They now sit on top of each other at the shoulder; you'll line them up in part 5.

Add it: select **knight_upperarm** and click **Add Component > Sprite Skin**. Click **Create Bones**. A bone appears as a child Transform of the part, called **Upperarm**. Do the same for the forearm and the hand: their bones are called **Forearm** and **Hand**.

Test it: select the Upperarm bone and rotate it with the **Rotate** tool (press E). The upper arm turns around the shoulder. Press Ctrl+Z.

5. Building the skeleton

Each part has its own bone, but they aren't joined yet.

1. Move the forearm so its top meets the bottom of the upper arm, and the hand so it meets the bottom of the forearm. Set **Order in Layer** on the Sprite Renderers so the parts overlap the way you want.
2. In the Hierarchy, drag the **Forearm** bone onto the **Upperarm** bone, and the **Hand** bone onto the **Forearm** bone. You now have a chain: shoulder, elbow, wrist.

Test it: rotate the **Upperarm** bone. The whole arm swings, forearm and hand included. Rotate the **Forearm** bone: only the forearm and hand bend. Press Ctrl+Z after each try. **Bones are just Transforms**: a child moves with its parent, and the Sprite Skin bends each sprite to follow its bone.

6. Animating the bones

1. Select **My Arm** and open **Window > Animation > Animation**. Click **Create**, save the clip in `Assets/_game/Animation/Clips/Knight` as `MyArm_Swing`. Unity adds an **Animator** and a controller to My Arm.
2. Click the red **Record** button. At frame 0, rotate the Upperarm bone to about Z **25**. Click further along the timeline (around 0:30), rotate it to about Z **−25**. At 1:00, set it back to 25. Keyframes appear as you go.
3. Click **Record** again to stop recording, and press Play in the Animation window to watch the swing.

Test it: press Play: your arm swings on a loop. Press Play again to stop.

7. IK: posing with a target

IK (inverse kinematics) works backwards: you say where the hand should be, and it works out the shoulder and elbow rotations.

Add it:

1. Select **My Arm**, **Add Component > IK Manager 2D**.
2. In the IK Manager 2D, click **+** and choose **Limb**. Unity adds a child object with a **Limb Solver 2D**.
3. On the Limb Solver 2D, drag the **Hand** bone from the Hierarchy into **Effector**. Click **Create Target**: a target object appears at the hand.

Test it: select the target and drag it around in the Scene view. The upper arm and forearm bend so the hand reaches it. Tick **Flip** on the Limb Solver 2D to bend the elbow the other way.

- A **Limb Solver** moves the two bones above the effector, like a real arm or leg.
- **IK overrides animated rotations** on the same bones, so your swing clip no longer moves the arm while the IK Manager is on. Untick the IK Manager 2D to get the swing back. IK is great for posing, and for keeping feet on uneven ground.

8. Compare with the finished knights

Open `CRE132_C13_SkeletalAnimation_Finished` (Unity Hub, **Add > Add project from disk**) and its scene `Assets/_game/Scenes/Core/C13_SkeletalAnimation`. Press **Play**.

- **Knight (Animated)**, in the middle, walks by rotating bones in its thighs, shins and arms, plus a small bob.
- **Knight (IK)**, on the right, has no animation. Open **IK Targets** inside it, select a target and drag it in the Scene view (it works in Edit mode too): the arm or leg bends to reach, just like yours.

Expand **Knight (Animated)** in the Hierarchy. It's built like your arm, with every part:

```
Knight (Animated)        Animator
|-- Sprites              one GameObject per part: Sprite Renderer + Sprite Skin
`-- Torso                bones (plain Transforms) in a hierarchy
    |-- Head
    |-- UpperArm Front > Forearm Front > Hand Front > Sword
    |-- UpperArm Back  > Forearm Back  > Hand Back
    |-- Thigh Front    > Shin Front    > Foot Front
    `-- Thigh Back     > Shin Back     > Foot Back
```

The same sprites are used for both arms and both legs. The far ones are tinted darker and drawn behind the torso with a lower **Order in Layer**.

Select **Knight (Animated)** and open **Window > Animation > Animation**. Pick `Knight_Walk` from the clip list: the legs swing about 25° each way, the shins bend as each leg passes, and the arms swing the opposite way, the same idea as your swing clip.

Open the **Animator** window (**Window > Animation > Animator**). Its controller has **Idle** and **Walk** states and a **Walking** bool, ticked by default. Press Play and untick **Walking** in the **Parameters** list: the knight goes back to idle.

Check your work

- ☐ Your arm has three parts, each with a **Sprite Skin** and a bone.
- ☐ The bones form a chain: rotating the upper arm bone swings the whole arm.
- ☐ Your `MyArm_Swing` clip swings the arm on a loop.
- ☐ With the IK Manager on, dragging the target bends the elbow; Flip bends it the other way.
- ☐ Compare with **Knight (Animated)** and **Knight (IK)** in `CRE132_C13_SkeletalAnimation_Finished` if something behaves differently.

Key points

- **Bones are just Transforms.** The Sprite Skin bends the sprite's mesh to follow them.
- **One bone per part** is like a puppet with pins. Several bones in one sprite, with weights, let a single picture bend: a cape, a tail or a long body.
- **IK sets rotations from a target position.** It overrides animated rotations on the same bones.
- **Skeletal suits big, smooth characters**; small pixel-art characters usually look better frame by frame.
- The **PSD Importer** package can import a layered `.psb` file as a whole character with all its parts already in place. That's the usual workflow when an artist gives you layered files.

If something goes wrong

- **There's no Create Bones button on the Sprite Skin:** the sprite has no bones. Check you dragged in a sprite from `KnightParts`, not another image.
- **Rotating the upper arm doesn't move the forearm:** the **Forearm** *bone* isn't a child of the **Upperarm** bone. Parenting the sprite GameObjects isn't enough.
- **The arm turns around the wrong point:** the bones were moved apart. Undo, or delete the part and drag it in again.
- **The IK target does nothing:** the **Effector** slot is empty, or the bones above the effector aren't joined in a chain.
- **My swing clip stopped working:** the IK Manager 2D overrides it. Untick the IK Manager 2D.

Try it

1. Make an **Attack** clip for your arm: rotate it back slowly, then forward fast.
2. Rig the other half: add `knight_thigh`, `knight_shin` and `knight_foot` as a leg, with its own **Limb** solver so you can pose the foot.
3. In the Skinning Editor, add a second bone to the torso and see how weights let one sprite bend. Click **Revert** afterwards: the bones belong to the `KnightParts` asset, so applying would change every part made from it, your arm included.